Chapter 5: PROPOSED SYSTEM

5.1 Overview

The proposed system, CareerOrbit, represents a departure from traditional linear learning models by implementing a multi-agent, hybrid intelligence architecture. This chapter details the structural and behavioral design of the platform, utilizing formal modeling techniques to illustrate how data transforms into cognitive insights.

5.2 High-End Multi-Tiered System Architecture

The CareerOrbit architecture is a five-layer stack designed to ensure pedagogical integrity and industrial scalability. As illustrated in **Fig 5.1**, the system transitions from raw user interactions to verified career outcomes.

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Parse error on line 2:
...ph TD    subgraph "Layer 5: Impact & Fu...
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'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
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'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
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'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE', 'THICK_ARROW_CROSS',
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'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'STR'
```

Fig 5.1: High-End Layered System Architecture of the CareerOrbit Ecosystem

5.2.1 Detailed Layer Analysis

Layer 1: Presentation (UX/UI)

The presentation layer is the primary entry point for the learner. It features a **Cinematic Architecture Canvas** that visualizes the curriculum as an interactive node-graph. The **Proactive AI Mentor** is integrated directly into this layer, utilizing Framer Motion for non-intrusive animations.

Layer 2: Persistence & Data (Neon DB)

This layer manages the "Digital Twin" of the learner. Unlike traditional databases, we utilize **JSONB Cognitive Memory** to store high-variance mastery probabilities for thousands of subtopics without schema rigidity.

Layer 3: Communication & API

This layer utilizes **Next.js Server Actions** to ensure zero-latency state updates. The **Real-time AI Streaming** allows the mentor to provide immediate guidance, while the **Secure Edge Middleware** handles authentication and data isolation.

Layer 4: Intelligence & Decision (The Cognitive Brain)

This is the core of CareerOrbit.

- **BKT Engine:** Performs the recursive Bayesian math to update the $pMastery$ vector.
- **Scaffolding Controller:** Dictates whether the LLM should provide a "Hint," a "Clarification," or a "Full Explanation" based on the BKT state.
- **Proactive Trigger Logic:** Monitors behavioral signals (e.g., idleness or repeated quiz failure) to automatically deploy the AI Mentor.

Layer 5: Impact & Fulfillment (Career Outcomes)

The final layer connects learning to employment. It maps BKT-verified mastery to **UN SDG 8 metrics** and facilitates a direct pipeline to **Job Portals**. Verified credentials are only generated once the BKT engine confirms the probability of mastery ($P(L)$) exceeds 0.95.

5.3 Use Case Modeling

To understand the functional boundaries of the system, we analyze the primary use cases for the learner. As illustrated in **Fig 5.3**, the user interacts with the system through four core lifecycle stages: Discovery, Diagnostics, Learning, and Mentorship.

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'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
'PE', '-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT',
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'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN', '==',
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'THICK_ARROW_OPEN', 'PIPE', 'STYLE', 'LINKSTYLE', 'CLASSDEF',
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'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'STR'
```

Fig 5.3: Use Case Diagram for the CareerOrbit Ecosystem

- **Discovery (UC1, UC2):** The initial phase where the user defines their vocational intent.

- **Diagnostics (UC3):** The BKT engine initializes the $pMastery$ baseline.
- **Learning (UC4, UC6):** The adaptive loop where mastery is tracked and visualized.
- **Mentorship (UC5):** The proactive support layer that fires during cognitive blockages.

5.4 Data Flow Analysis

The flow of data within CareerOrbit is characterized by the transformation of raw behavioral signals into refined mastery probabilities.

5.4.1 DFD Level 0 (Context Diagram)

Fig 5.4 illustrates the context diagram, showing CareerOrbit as a central process interacting with external entities like the OpenAI API for reasoning and the Neon Cloud for data persistence.

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...B Snapshots" --> DB[(Neon Database)]
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'PCT', 'COMMA', 'ALPHA', 'COLON', 'BRKT', 'DOT', 'PUNCTUATION',
'UNICODE_TEXT', 'PLUS', 'EQUALS', 'MULT', got 'PS'
```

Fig 5.4: DFD Level 0 — Context Diagram

5.4.2 DFD Level 1 (Process Decomposition)

In **Fig 5.5**, we decompose the central system into its constituent processes. This diagram highlights the "BKT-LLM Feedback Loop," where the output of the BKT process directly modifies the inputs for the LLM process.

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flowchart TB    U[U
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Fig 5.5: DFD Level 1 — Process Decomposition showing the BKT-LLM feedback loop

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5.5 Cognitive State Transition Modeling

To mathematically formalize the adaptive behavior of CareerOrbit, we model the Knowledge Component (KC) lifecycle as a **Hidden Markov Model (HMM)**. As illustrated in **Fig 5.6**, the learner exists in one of two hidden states: **Unlearned** or **Learned**.

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stateDiagram-v2
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Fig 5.6: State Transition Diagram representing the Bayesian Knowledge Tracing HMM

In **Fig 5.6**, we observe that mastery is treated as a permanent transition. The system's primary task is to use the **Observations** (Correct/Incorrect) to estimate the probability of the user being in the **Learned** state.

- **The Learning Transition ($P(T)$):** Represents the cognitive shift after an instructional intervention (e.g., watching a video).
- **The Persistence Invariant:** Once a user reaches the "Learned" state, the model assumes a probability of 1.0 for staying in that state, although **Asymptotic Clamping** is applied in the implementation to maintain model responsiveness to "Slips."

5.6 System Sequence Analysis

The temporal interaction between the system's components is critical for achieving low-latency adaptation. As depicted in the sequence diagram in **Fig 5.7**, the "Adaptive

Loop" is a five-stage process involving the client, the serverless engine, the database, and the LLM inference service.

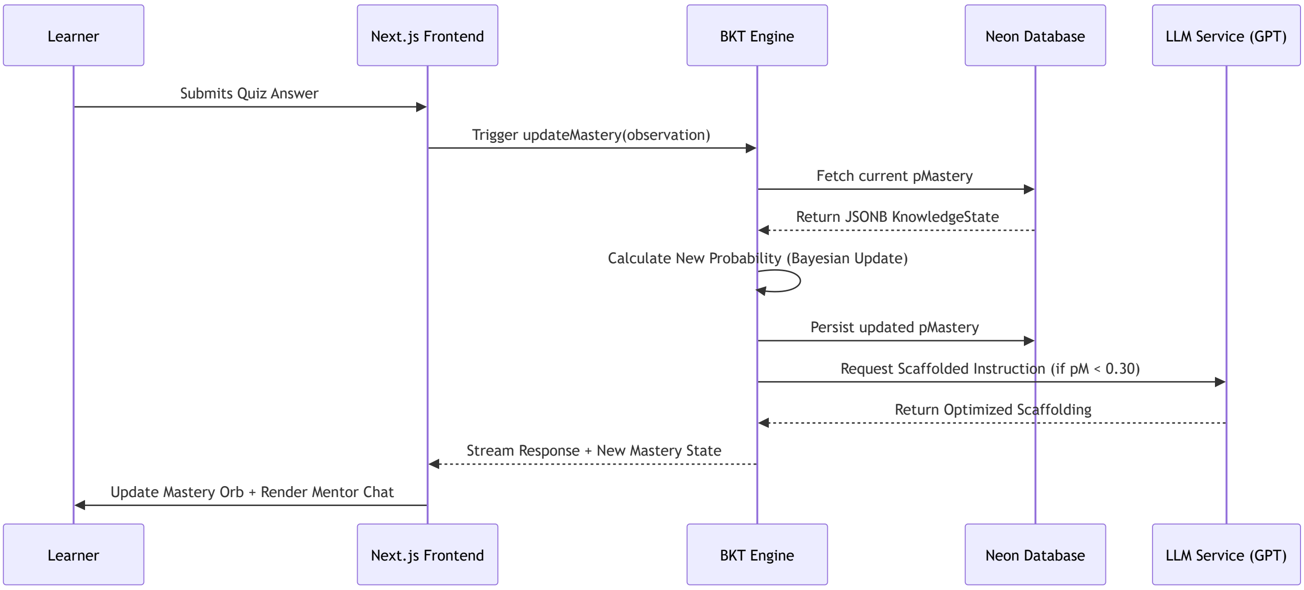

Fig 5.7: Sequence Diagram of the Real-time Adaptive Learning Loop

As shown in **Fig 5.7**, the BKT engine acts as the central orchestrator. It ensures that the LLM is only invoked *after* the deterministic state update is performed, ensuring that the guidance provided is always grounded in the most recent evidence of user ability.

5.7 Entity-Relationship Modeling (ERD)

The structural integrity of the CareerOrbit ecosystem is maintained through a relational schema optimized for both structured and semi-structured (JSONB) data. **Fig 5.8** illustrates the ER Diagram, highlighting the central role of the `user_model` table.

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erDiagram    USERS
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Fig 5.8: Entity-Relationship Diagram (ERD) for CareerOrbit Cognitive Data Persistence

5.7.1 Data Modeling Rationale

As seen in **Fig 5.8**, we utilize a hybrid storage strategy:

1. **Relational Columns:** Used for high-frequency queries (e.g., `userId`, `status`).
2. **JSONB Columns:** Used for high-variance cognitive data (`knowledgeState`) and hierarchical content (`modules`). This ensures that the schema remains flexible as new vocational paths are added to the system without requiring invasive migrations.

5.8 Summary

Chapter 5 has provided a comprehensive architectural and behavioral analysis of the CareerOrbit platform. By detailing the high-level architecture (**Fig 5.1**), the process-level data flow (**Fig 5.5**), and the underlying cognitive state machine (**Fig 5.6**), we have demonstrated that the system is not merely a collection of features, but a mathematically consistent learning engine. The structural design depicted in the ERD (**Fig 5.8**) further ensures that the platform is scalable, secure, and research-ready.